INCH-POUND
MIL-DTL-24640/15E
8 November 2011
SUPERSEDING
MIL-DTL-24640/15D
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRIC, -20 TO +105 °C, 600 VOLTS, LIGHTWEIGHT, MULTI-PAIR, 22 AWG, TYPE 2XSAW (INCLUDING VARIATION TYPES 2XSAOW AND 2XSAWA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24640.

Construction (Watertight)

First	-	Tin-coated copper conductor in accordance with table I.
Second	-	Insulation material and basic construction requirements shall be in accordance with SAE-AS81044/12 and tested in accordance with the component wire ratings and requirements listed on this specification sheet. Conformance testing shall be in accordance with SAE-AS81044 prior to cabling.
Third	-	Two conductors, one black and one white, twisted together. The pitch shall be 1¼±¼ inch.
Fourth	-	Optional binder tape.
Fifth	-	Braided shield of 38 AWG tin-coated copper strands. Braid coverage shall be no less than 90 percent determined in accordance with WC-27500.
Sixth	-	Shield insulation of two polyester tapes or a polyester tape and extruded jacket. Identification codes applied using Method 2.
Seventh	-	The required number of shielded, twisted pairs shall be cabled together. Cabling sequence shall be consecutive, starting with number 1 from center outward. Fillers may be used, as necessary, to form a round cable.
Eighth	-	Optional binder tape.

Construction Variant for Type 2XSAW (see table II)

Ninth	-	Cross-linked polyolefin jacket. NOTE: Manufacturer's identification tape or thread is required (see 3.7.3).

Construction Variant for Type 2XSAOW (see table IV)

Ninth	-	Braided shield, tin-coated copper strands. Braid must be constructed to meet surface transfer impedance requirements as listed herein.
Tenth	-	Optional binder tape.

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Eleventh - Cross-linked polyolefin jacket.
NOTE: Manufacturer's identification tape or thread is required (see 3.7.3).

Construction Variant for Type 2XSAWA (see table III)

Ninth - Cross-linked polyolefin jacket.

Tenth - Braided metal armor.
NOTE: Manufacturer's identification tape or thread is required (see 3.7.3).

TABLE I. Component wire details.

Watertight conductor			Finished wire			
Wire size (AWG)	Stranding (no. x AWG)	Diameter nominal (inches)	Diameter (inches)	Test load life cycle accelerated aging tests (lbs ±3%)	Low temp. at -40 °C	
					Mandrel diameter (inches)	Test load (lbs)
22	7 x 30	0.030	0.047 ±0.003	0.190	0.75	1.0

TABLE II. Construction details (unarmored cable).

PIN M24640/15	Type number	Number of pairs	Jacket thickness (inch) (min.)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max.)
				(min.)	(max.)	
-01UN	2XSAW-3	3	0.041	0.368	0.396	115.0
-02UN	2XSAW-7	7	0.041	0.461	0.497	195.0
-03UN	2XSAW-14	14	0.041	0.641	0.691	390.0

TABLE III. Construction details (armored cable).

PIN M24640/15	Type number	Number of pairs	Jacket thickness (inch) (min.)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max.)
				(min.)	(max.)	
-01AN	2XSAWA-3	3	0.041	0.418	0.456	148.0
-02AN	2XSAWA-7	7	0.041	0.511	0.557	244.0
-03AN	2XSAWA-14	14	0.041	0.691	0.751	446.0

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TABLE IV. Construction details - overall shield.

PIN M24640/15	Type number	Number of pairs	Jacket thickness (inch) (min.)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max.)
				(min.)	(max.)	
-01UO	2XSAOW-3	3	0.041	0.405	0.437	160.0
-02UO	2XSAOW-7	7	0.044	0.510	0.550	260.0
-03UO	2XSAOW-10	10	0.044	0.631	0.681	385.0
-04UO	2XSAOW-14	14	0.047	0.689	0.743	465.0
-05UO	2XSAOW-19	19	0.047	0.757	0.817	627.0
-06UO	2XSAOW-24	24	0.047	0.884	0.952	800.0
-07UO	2XSAOW-30	30	0.050	0.941	1.02	910.0
-08UO	2XSAOW-37	37	0.050	1.01	1.09	1050.0

REQUIREMENTS:

Qualification required.

Construction details: See tables I, II, and III.

Component wire: The component wire shall be in accordance with SAE-AS81044 and shall be tested in accordance with the following wire ratings and additional requirements.

Temperature rating	135 °C (275 °F) (max.) conductor temperature
Voltage rating	600 volts (root mean square) at sea level
Accelerated aging	Oven temperature, 250±2 °C (482±3.6 °F) for 6 hours
Low temp. cold bend	-40 °C (-40 °F) for 4 hours
Blocking	150±2 °C (302±3.6 °F)
Color	In accordance with MIL-STD-104, Class 1
Flammability	30 seconds (max.)
	3.0 inches (max.)
	No flaming of tissue paper
Humidity resistance	5000 megohms for 1000 feet (min.) insulation resistance after humidity exposure
Dielectric test	
Primary insulation (when tested in lieu of spark test)	6.0 kilovolts (peak), 100 percent impulse test
Finished wire	8.0 kilovolts (peak), or 3 kHz spark test 5.5 kV, 100 percent test
Insulation resistance	5000 megohms for 1000 feet (min.)
Life cycle	Oven temperature, 200±2 °C (392±3.6 °F) for 120 hours

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Physical properties of primary insulation	
Tensile strength	2500 lb/in^2 (min.)
Elongation	150 percent (min.)
Shrinkage	0.125 inch (max.) at 250±2 °C (482±2 °F)
Spark test primary insulation	1500 volts (root mean square), 3 KHz, 100 percent test
Surface resistance	500 megohms inches (min.), initial and final readings
Thermal shock	Oven temperature, 150±2 °C (302±3.6 °F)
Maximum change in measurement	0.060 inch
Wet dielectric test	2500 volts (root mean square)
Wrap test	Mandrel wrap test required

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	17.89
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor and shield	2000
Conductor to shield	1000
Component shield to component shield	200
Component shield to overall shield (2XSAOW only)	500
Insulation resistance (megohms, 1000 feet, min.)	
Conductor to conductor and shield	2500
Shield to shield (component shield only)	100
Shield to shield (component shield to overall shield, 2XSAOW only)	100
Conductor and shield continuity	No failure
Jacket flaws (Type 2XSAOW only)	No failure

Group A:

Visual and dimensional	No failure
Capacitance:	
Mutual capacitance (at 1 MHz, picofarads/foot (pF/ft), max.)	55
Capacitance unbalance (percent, max.)	15
Characteristic impedance (at 1 MHz, ohms)	33 (min.), 72 (max.)
Watertightness (see table IX of MIL-DTL-24640 for limits of water leakage)	No failure
Weight	No failure

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Group B:	
Cold bending	No failure
Drip	No failure
Cross-link proof test (percent, max.)	
Jacket	50
Tear (lb/in thickness, min.)	35
Physicals (unaged)	
Jacket	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Attenuation (at 3 MHz, dB/100 feet, max.)	Not applicable
Gas flame (1 hour)	Not applicable
Group C:	
Armor (conformance to material, construction and coverage)	No failure
Physicals (aged), air oven	
Jacket	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (conductor insulation, cycles, min.)	Not applicable
Permanence of printing (jacket, cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable sealant removability	No failure
Shrinkage	No failure
Surface transfer impedance (Type 2XSAOW only)	
Milliohms per meter (max.)	700
EMP response (dB, min.)	60
Group D:	
Flame propagation	No failure

Qualification Inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable)	No failure
Armor	No failure

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Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	10
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	Not applicable
Component wire	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)	
2XSAW-3, 2XSAW-7, 2XSAW-14, 2XSAOW-3, 2XSAOW-7, 2XSAOW-10, 2XSAOW-14, 2XSAOW-19, 2XSAOW-24, 2XSAOW-30, 2XSAOW-37	3 & 14	500
	19 - 37	250

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2009-083)

Review Activities:
Army – AR, AV, CR, CR4
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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